CON201

Orchestrating containers on AWS

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Microservices and containers

AWS container services

Datree story and live

demo

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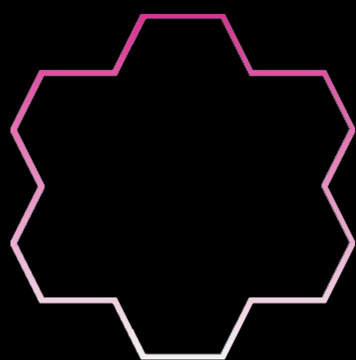
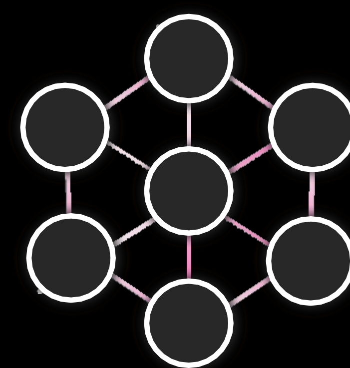
When the impact of change is small, release velocity can increase



Monolith
Does everything



Microservices
Does one thing

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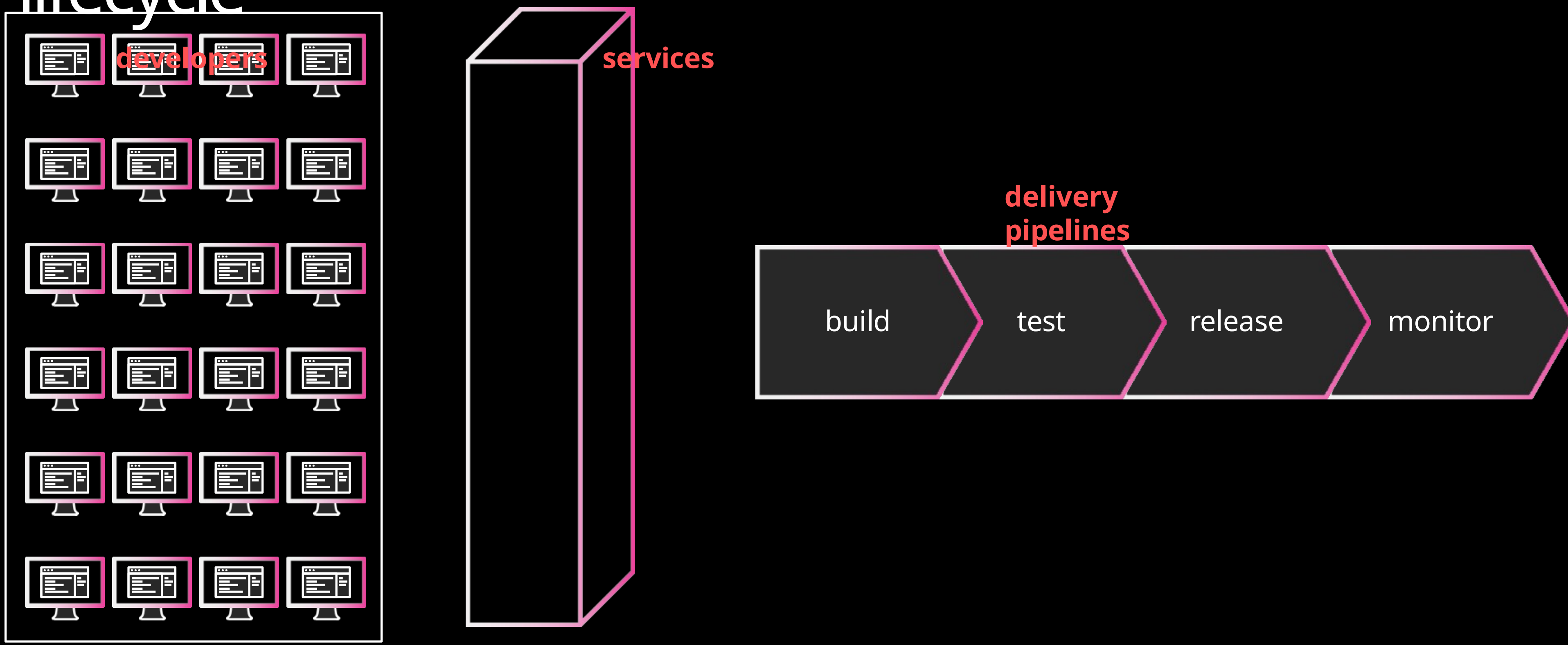
Monolith development lifecycle



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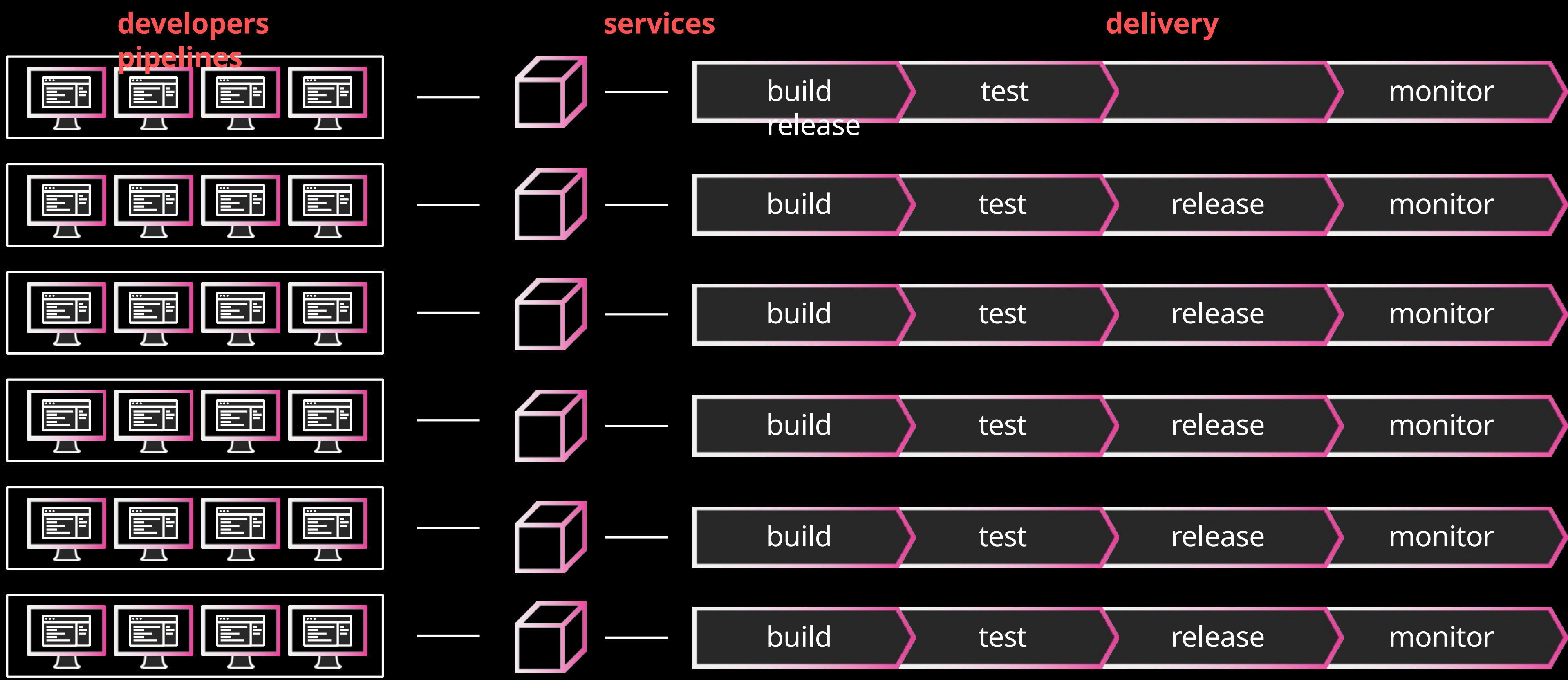
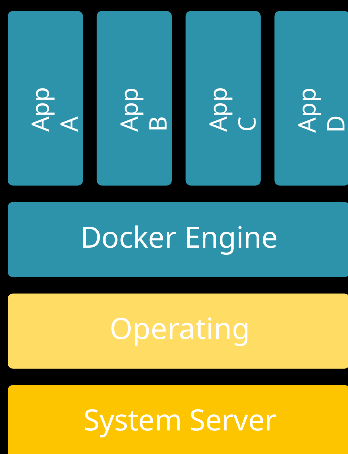
Microservice development lifecycle



Containers and Docker

A container is a standard unit of software that packages up code and all its dependencies so the application runs quickly and reliably from one computing environment to another.[1]



[1] https://www.docker.com/resources/what-container

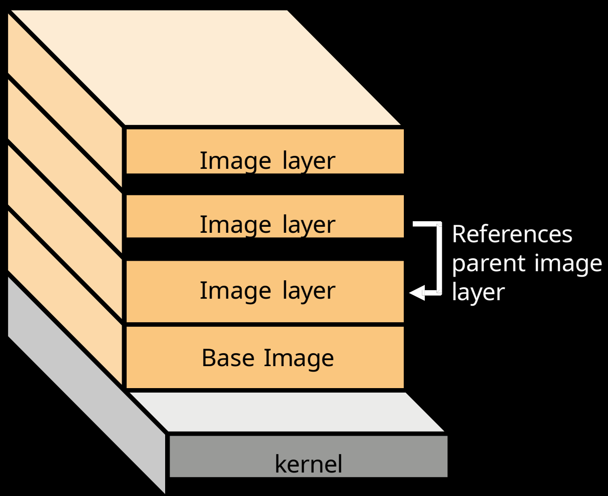
Docker Image

Used to launch container

Instructions documented in Dockerfile

Merge layers into single image

Read-only template



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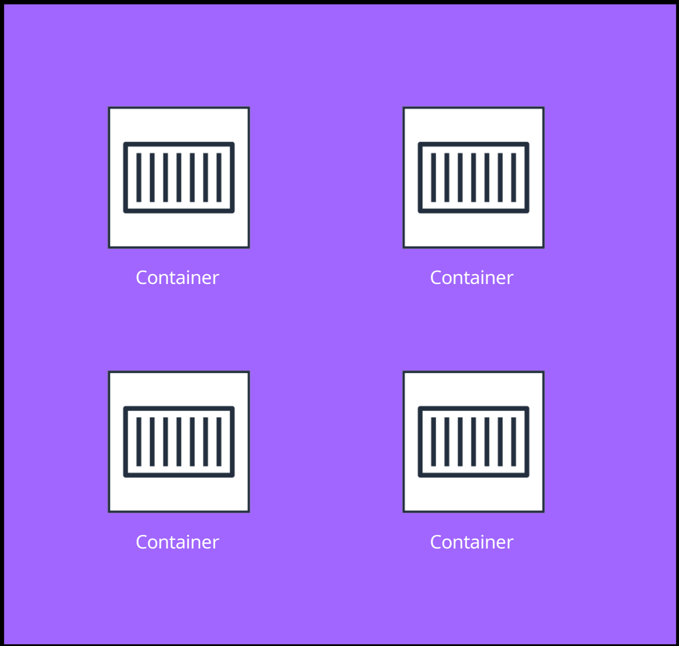
Containers and Microservices

- Do one thing, really well
- Any app, any language
- Isolated execution environment
- Test and deploy same artifact
- Faster startup



Containers have become the standard for how to ship and run your application in the cloud

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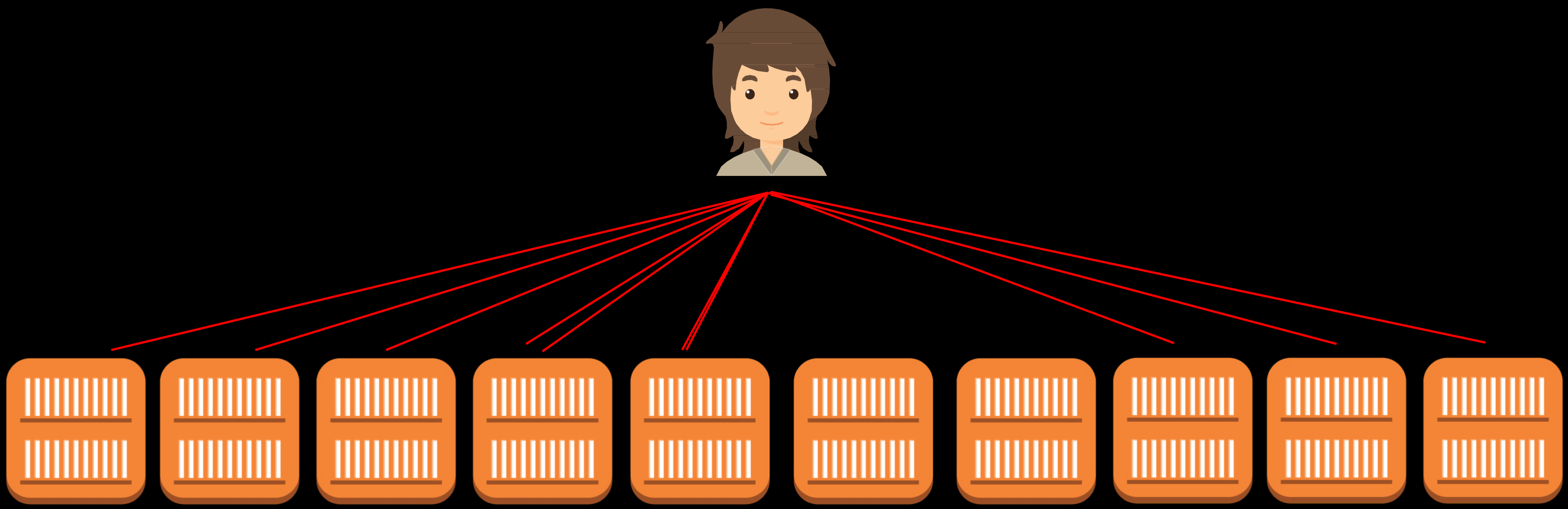
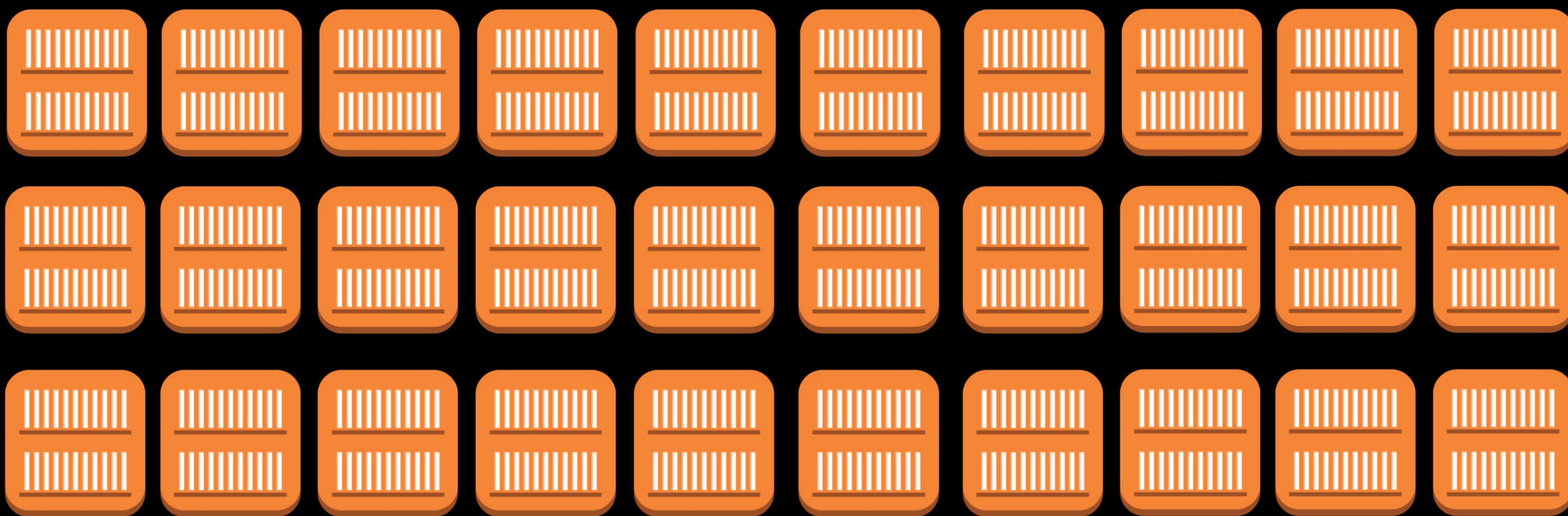
Manually downloading and launching containers by hand is inefficient and error prone



Container orchestration



AWS container services

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AWS Container Services landscape

MANAGEMENT

Deployment, scheduling, scaling & management of containerized applications

Amazon ECS

Amazon EKS

HOSTING

Where the containers run

Amazon EC2

AWS Fargate

IMAGE REGISTRY

Container image repository

Amazon ECR

Kubernetes

Open source container management platform

Helps you run containers at scale

Gives you primitives for building modern applications

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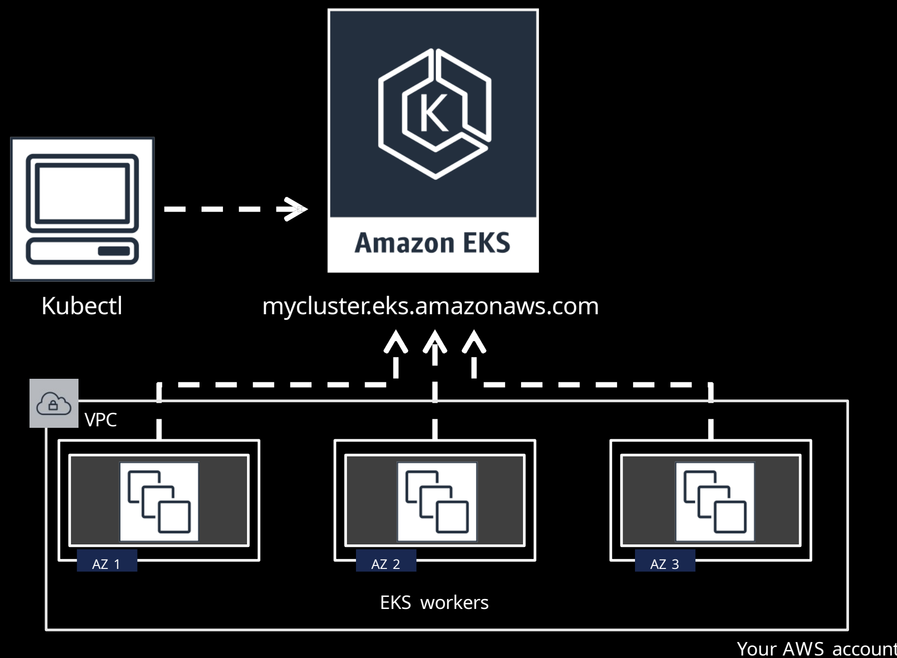
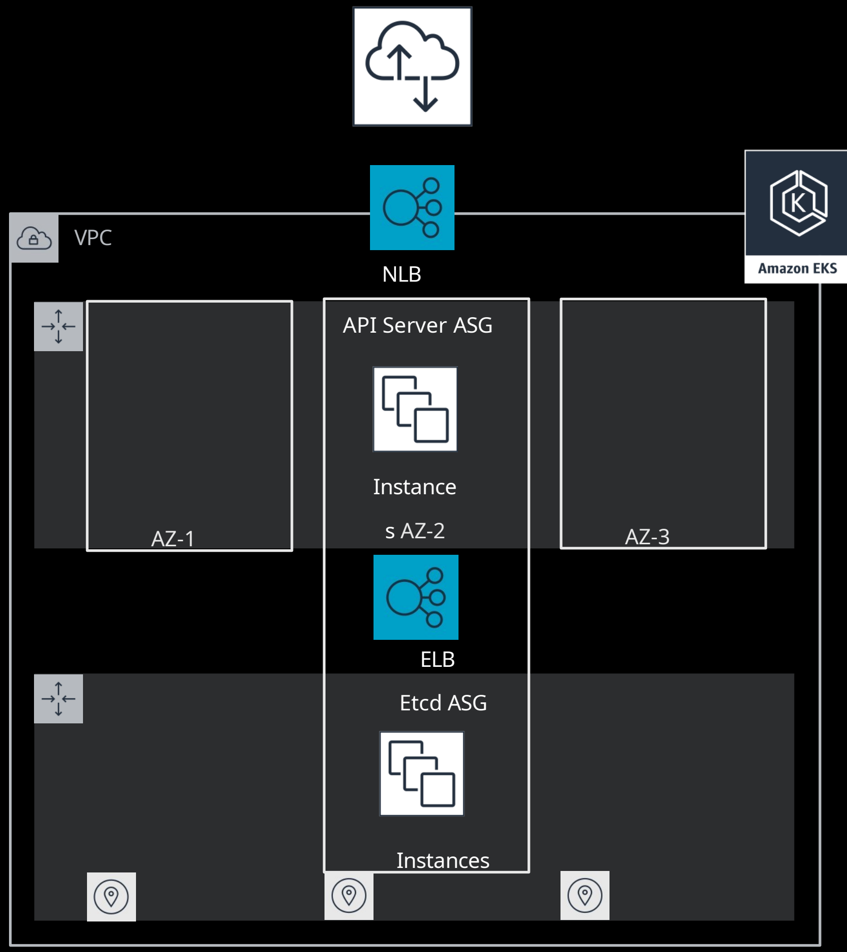
Amazon EKS architecture



Kubernetes control plane

Highly available and single tenant infrastructure

All “native AWS” components
Fronted by an NLB



Amazon EKS features

Certified conformant

Integration with Elastic Load

Balancing Managed updates

IAM authentication

Amazon Elastic Container Service

ECS

Highly Scalable , Highly Performant Container Management System

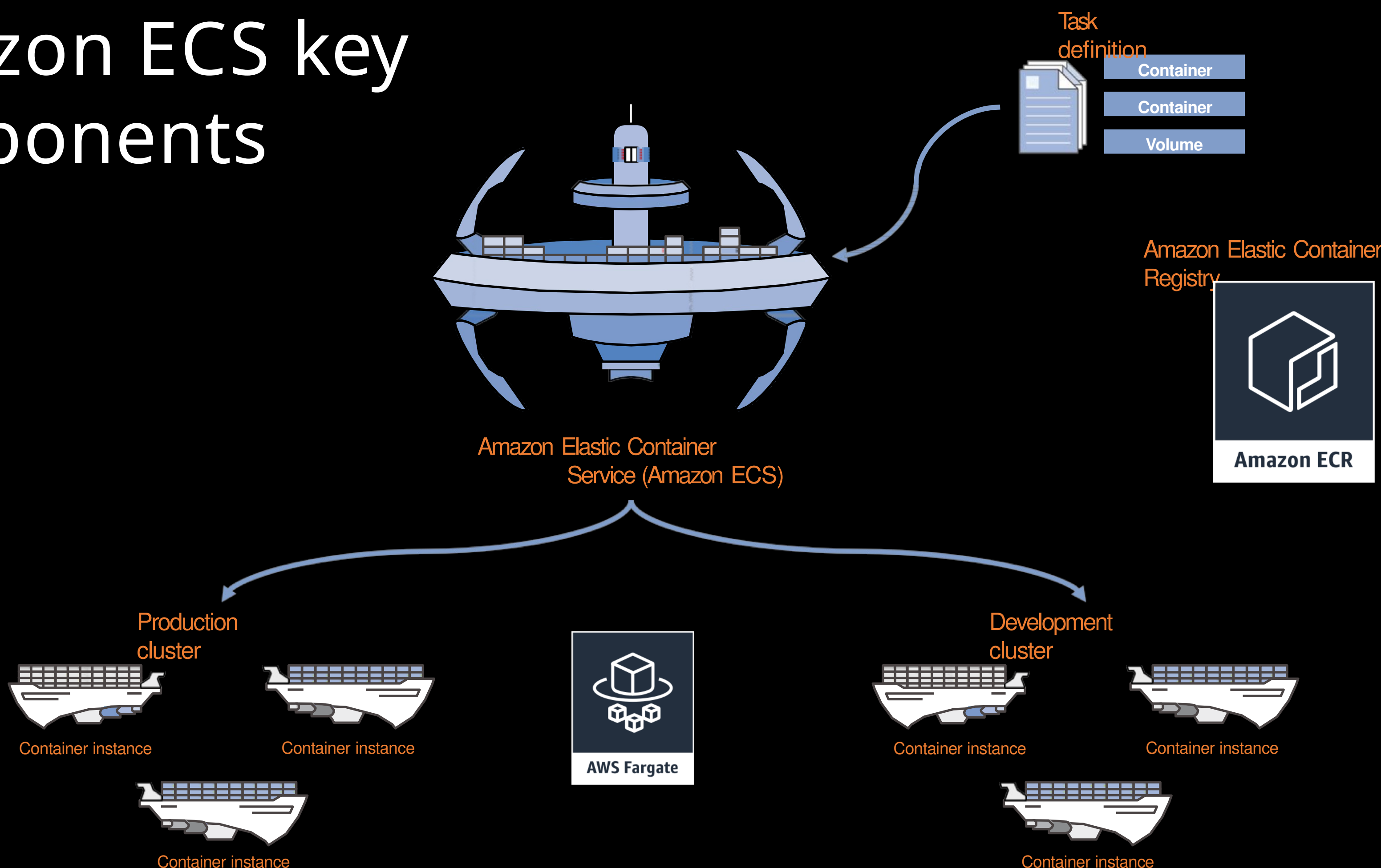
A managed platform

Cluster Manageme
nt

Container Orchestration
& Placement

Deep AWS Integratio
n

Amazon ECS key components



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Amazon ECS features

Integration with Elastic Load Balancing

Service Discovery with AWS

CloudMap Task level IAM support

Blue/Green Deployments with AWS

CodeDeploy Windows Containers

Compatibility

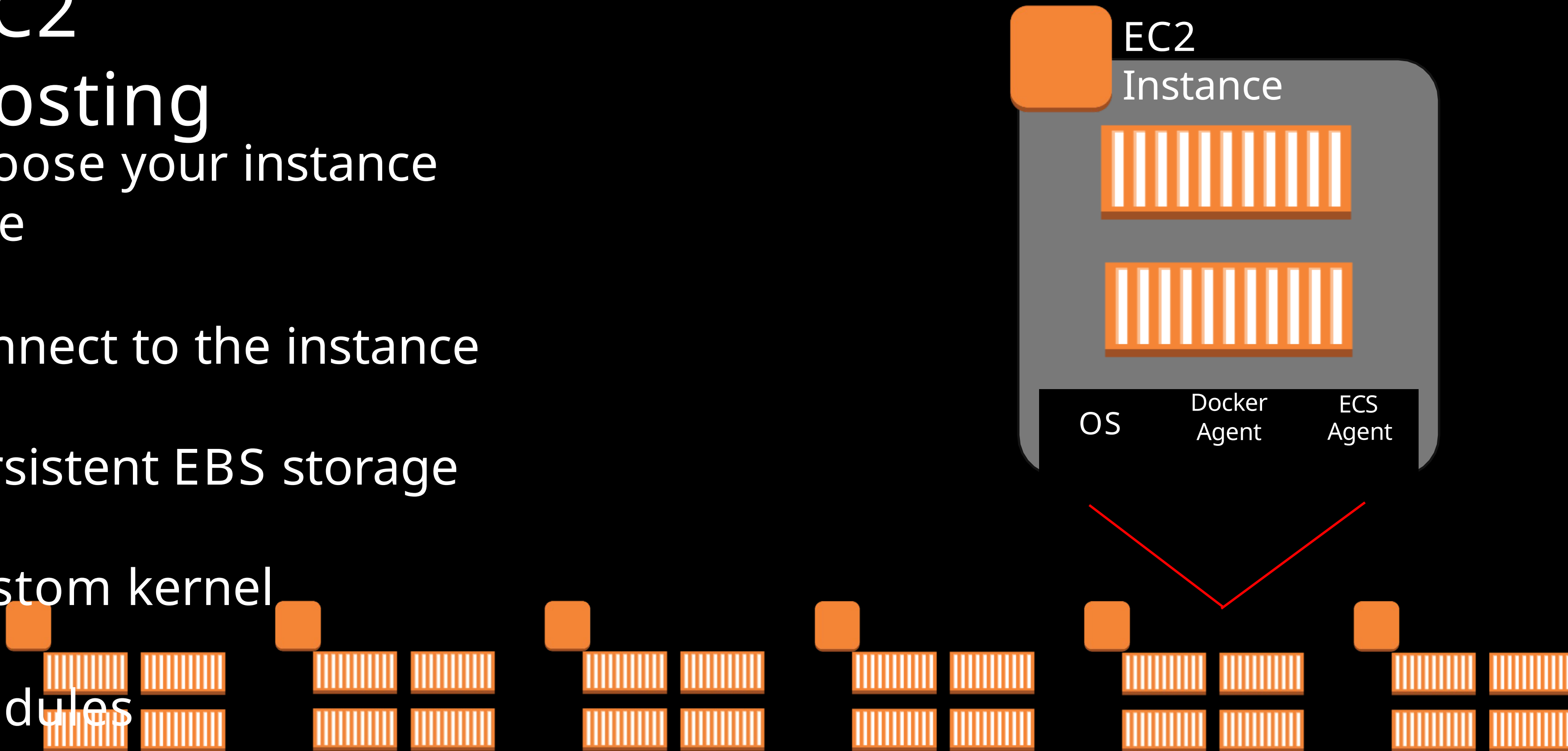
EC2 hosting

Choose your instance type

Connect to the instance

Persistent EBS storage

Custom kernel modules



"Just launch 10 copies of my container distributed across three availability zones and connect them to this load balancer"

Shimon Tolts
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1. About me & datree.io
2. What is Fargate TCO?
3. Why did we choose to use Fargate
4. How do we deploy (technical)
5. DEM O

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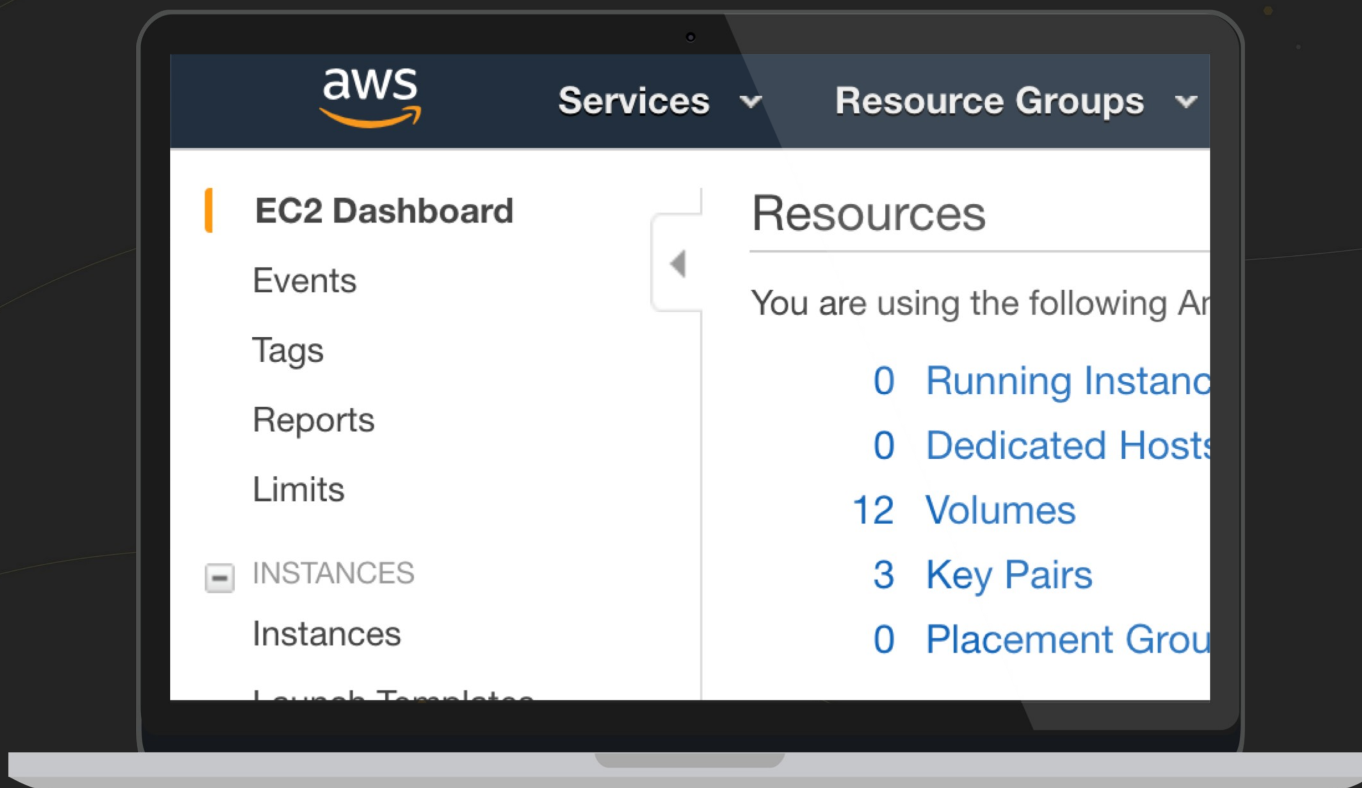
About me


```
Shimon Tolts {
  age: 30,
title: “CTO & Co-Founder @ datree.io”,
  misc: [“AWS Community Hero”,“Gamer”]
}
```

0 EC2 instances @ datree.io



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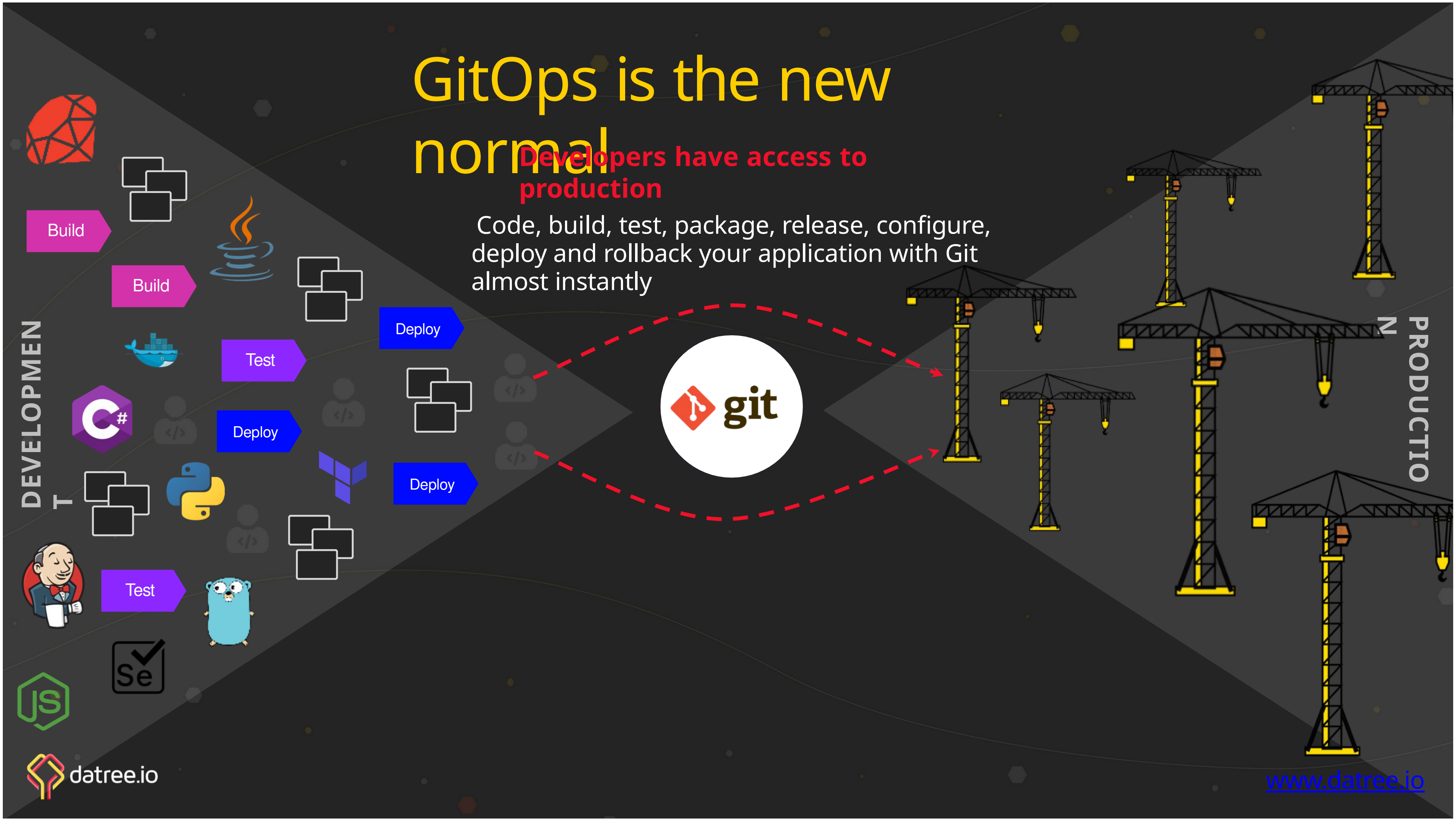
GitOps

Datree is a Policy Enforcement Platform for confident and compliant code.


GitOps is the new normal
Developers have access to production
Code, build, test, package, release, configure, deploy and rollback your application with Git almost instantly
DEVELOPMENT
Build
Build
Test
Deploy
Test
Deploy
Deploy
git
PRODUCTION
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Datree connects with GitHub

Datree connects with **GitHub** to provide automatic policy compliance checks and insights for every code commit and pull request.

Datree connects with GitHub

Set Code Policies

Create custom policies or choose from recommended defaults, and choose where they should be enforced.

- Separate secret credentials from source code
- Include .gitignore in every project
- Link pull request title to a Jira ticket
- Create custom policies...

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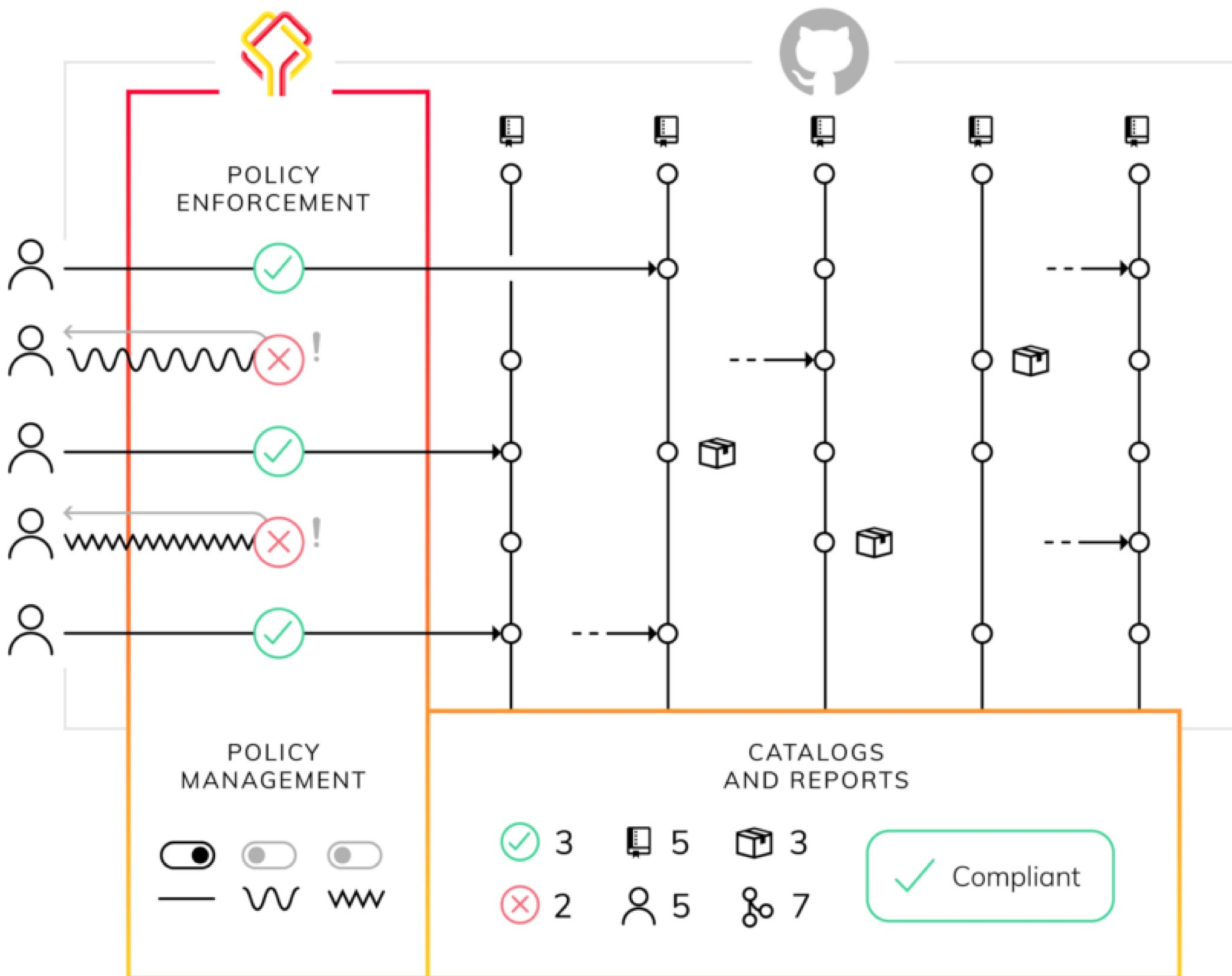
POLICY
ENFORCEMENT
POLICY
MANAGEMENT
CATALOGS
AND REPORTS
3
5
3
2
5
7
Compliant

Apples and oranges

AWS Fargate != Amazo
n EC2

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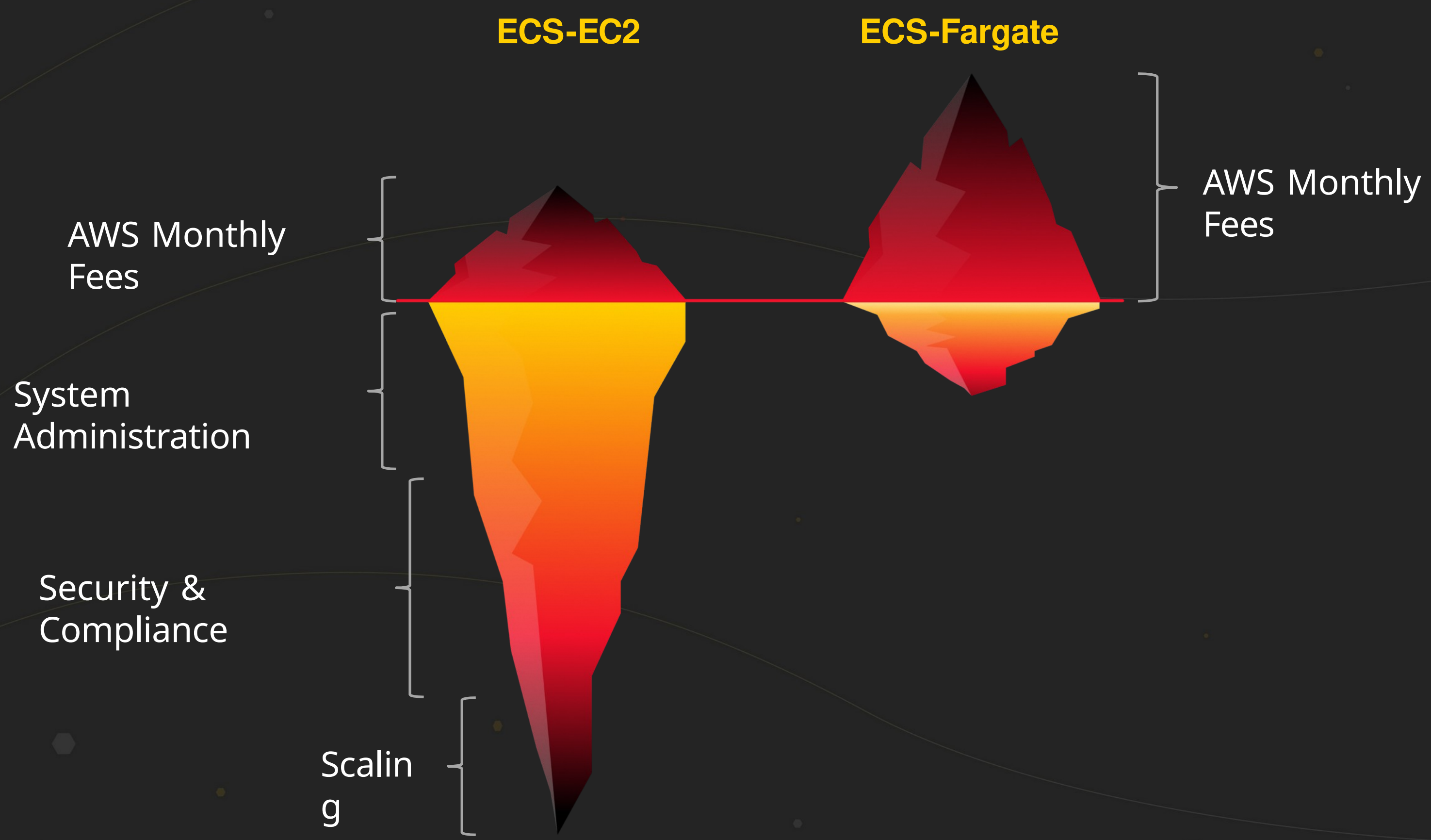
Ops benefits



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No more Amazon EC2 management

We no longer configure AMIs

Monitoring & logging is built in

Operating system management

All of our code is packaged using Docker containers, so we are **ONLY** responsible for what runs within our containers

No more:

- Linux Patching
- Docker service updates
- ECS Agent updates

Scaling

We no longer deal with scaling EC2 fleets

Taking care of bin packing our instance to run cost-effectively

Compliance & security

Compliance and security are our top priorities

Out-of-the-box security as a

service.Fargate comes

certified with

SOC 2 HIPAA PCI-DSS

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Building an icecream service

- http://icecream.datree.io
- A simple Node.js app
- A web service using Koa.js
- Serving icecream!

Pipeline overview

- Source code in GitHub
- AWS Application Load Balancer
- AWS ECS Fargate cluster
- GitHub Actions workflow (CI/CD)

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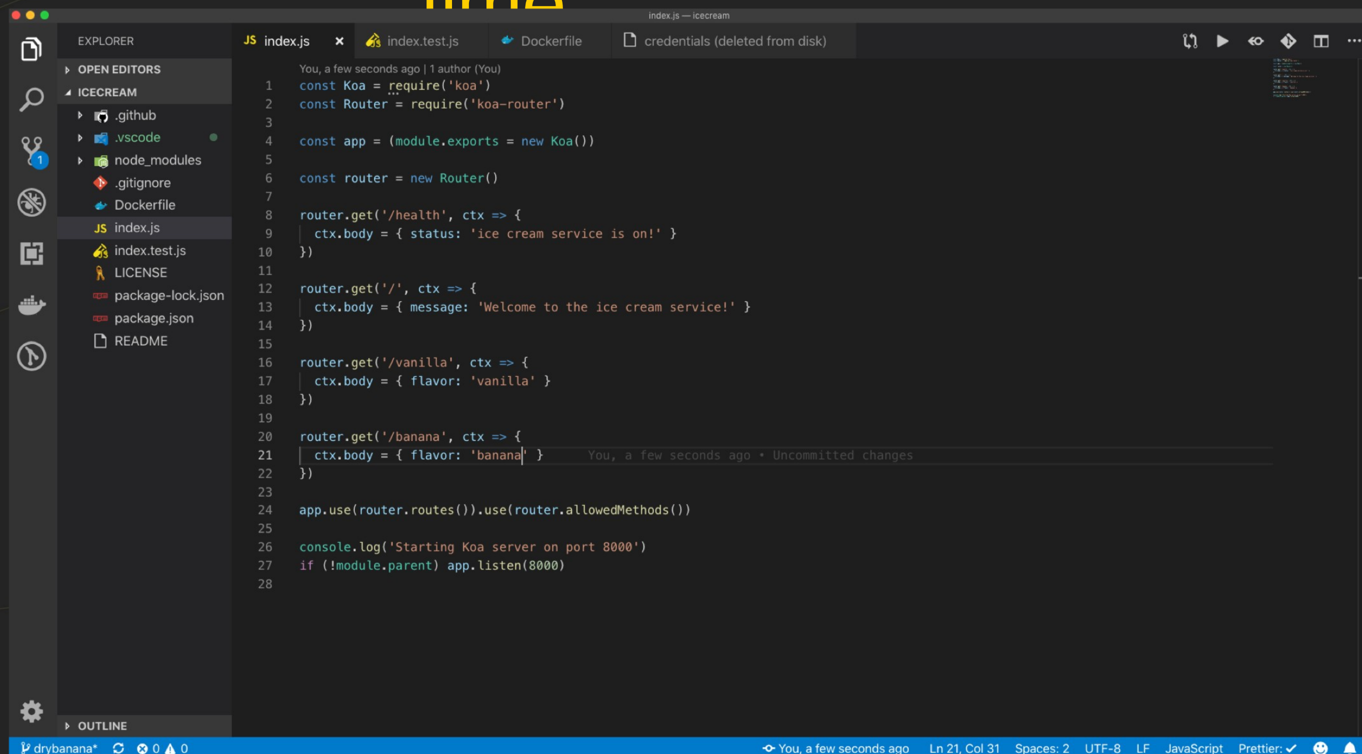
DEMO time



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ECS on EC2 vs. Fargate

- 10 GB disk space limit
- No instance type selection (GPU/CPU/MEM optimized)
- Amazon EBS attaching is not available
- No Reserved Instance pricing
- No Spot Instances support

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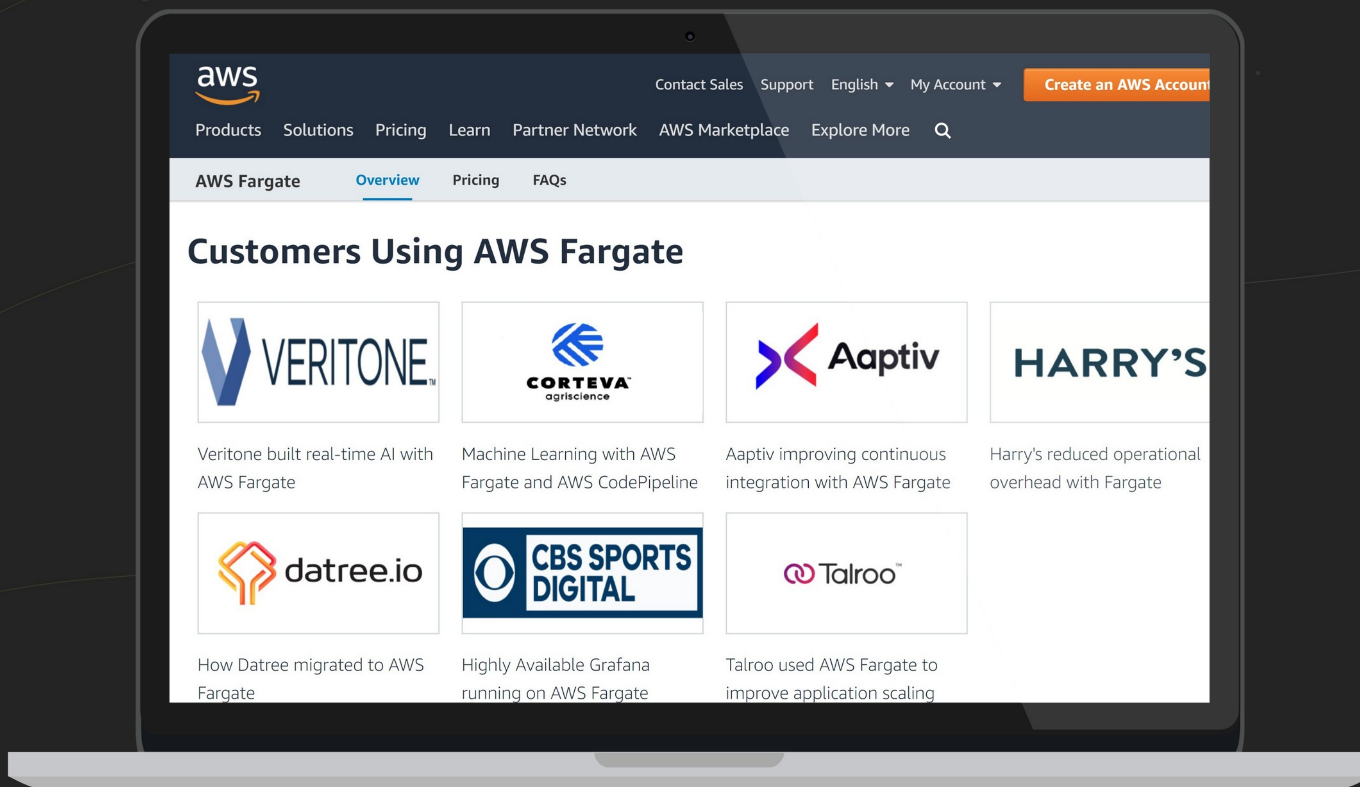
Open case study on AWS Fargate page



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Resources

1. https://datree.io/blog/migrating-to-aws-ecs-fargate-in-production/

2. https://www.youtube.com/watch?v=rtk3rRdAZ6s&feature=youtu.be &t=1239

3. https://github.com/silinternational/ecs-deploy

Related breakouts

14:10 Deep Dive on Amazon Elastic Container Service (ECS) Brent Langston

15:00 Mastering Amazon Elastic Container Service for Kubernetes (Amazon EKS) Kobi Biton, Chen Fisher

15:50 From Code to a running container Alexei Ledenev, Gal Marder

Thank you!

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Dima Breydo

Shimon Tolts

http://bit.ly/2SJ6Md2

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